

Musanze Safe Markets

Field Inspection Prototype

Group number **03** · Group verification code **MOB-G03-7014**

Group leader: Mohammed Osama Hasan, 25/27014. The code is group 03 plus the last four digits of that registration number. The same code must appear in the app header and on the review screen.

1. Cover and scenario

Role	Name	Registration number
Member 1 — Product and UX lead	Nuzha ZainEl-Abdeen Mohammed Ismail	25/27419
Member 2 — Interface engineer	Ehab Fakhralden Mohamed Hamid	25/27950
Member 3 — State and navigation engineer	Mojtaba Abdalitieef Ahmed	25/27660
Member 4 — Device integration and QA lead. Group leader.	Mohammed Osama Hasan	25/27014
Member 5 — Release and evidence lead	Lazarus Simboya Ira Inyasio	25/28180

Problem statement

Musanze Safe Markets is running a one-day pilot. A field officer must see the assigned market zones, record one fictional vendor inspection, and attach one evidence photo. There is no server. Names and phone numbers in the demo are fictional. The record lives only in the phone’s memory for this session.

Target field officer

Primary user: a market field officer walking between stalls with one hand free, often in bright sun, using Expo Go on an Android phone. They are not a designer. They need the next step to be obvious, errors to say what to fix, and the group code visible so the assessor can confirm the right build.

What this document owns

Member 1 defines the scenario, the task flow, the route map, the wireframes, the visual system, and the accessibility rules. Members 2, 3, and 4 implement those rules. The built app must match this document.

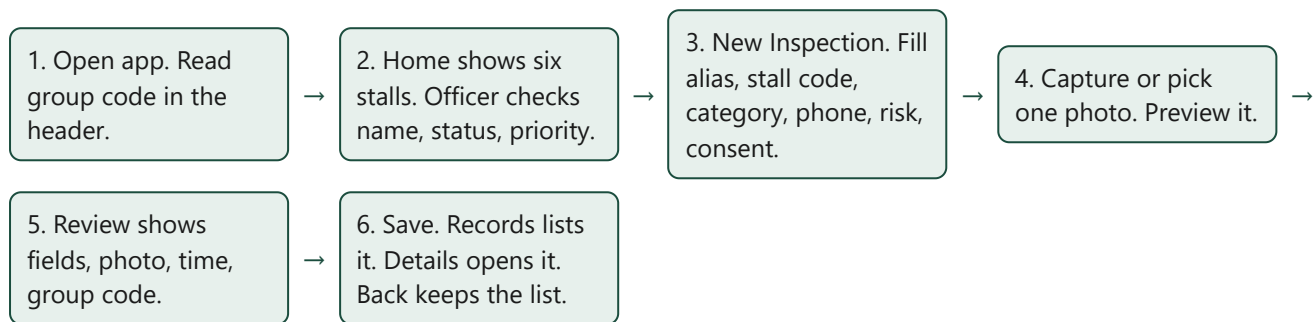
2. User and task flow

Primary user. Field officer on the Musanze Safe Markets pilot.

Goal. Register one fictional stall inspection and keep it in an in-memory list that can be opened again from Records.

Constraints. Expo Go on Android. No backend, database, API key, or real personal data. Larger system font and a narrow screen must stay usable. Invalid input must not reach the saved list.

Happy path



Alternate paths the assessor will try

- **Empty catalog.** The officer types a search that matches nothing. The list is replaced by an empty state with a way to clear the search.
- **Invalid form.** Blank fields, a stall code that breaks the pattern, a phone that is not a fictional Rwanda mobile number, or missing consent. Each fault shows a message under that field. Review does not open.
- **Permission denied.** The camera or gallery permission is refused. The form stays. A short message explains what happened and offers Try again. The other source (camera or gallery) remains available.
- **Picker cancelled.** The officer backs out of the camera or gallery. Message: "No image selected. You can try again." Nothing else is cleared.
- **Replace or remove.** After a preview, Replace opens the source again. Remove clears the preview and blocks Review until a new image is attached.
- **Back.** From Details, back returns to Records with the same in-memory list. Switching tabs does not wipe the form or the saved records.

3. Information architecture

Bottom tabs are the three jobs. A stack sits above the tabs so Details and Review can be pushed and popped without losing the session.

Route name	Where	What is passed
MainTabs	Root stack, first screen	None. Reads the in-memory session.
Home	Tab	None. Reads the fixed catalog of six zones.
NewInspection	Tab	None. Local form state. On success, navigates to Review with the draft.
Records	Tab	None. Reads the in-memory inspection list.
Review	Stack screen	InspectionDraft: all validated fields plus image URL.
InspectionDetail	Stack screen from Records	inspectionId: string. The screen looks up the record in memory.

Catalog record (fixed, fictional)

Name	Category	Status	Priority
Kinigi Produce Row	Produce	Open	High
Muhoza Grain Shed	Grains	Pending	Medium
Cyuve Meat Corner	Meat	Flagged	High
Busogo Clothing Lane	Clothing	Open	Low
Kimonyi Household Bay	Household	Closed	Low
Musanze Cooked Food	Cooked food	Pending	Medium

Every card also has a grey image placeholder with a short text label, for example “Produce stalls”. The catalog is not editable.

Inspection the officer types

- **Vendor alias** — required, at least 2 characters. Example: Mama Keza.
- **Stall code** — required, exact pattern **MZ-[A-F]-NNN**. Example: MZ-A-014. Letters A to F only. Three digits.
- **Category** — required, one of Produce, Grains, Meat, Clothing, Household, Cooked food.
- **Contact number** — required, fictional Rwanda mobile: **+250 7XX XXX XXX**. Example: +250 788 123 456.
- **Risk level** — required, Low, Medium, or High.
- **Consent** — required checkbox: “I confirm this record uses fictional demo data only.”
- **Evidence image** — required before Review. One photo from the camera or the gallery.

On Save, the app adds an id and an ISO timestamp and appends the record to the in-memory list. Leaving the app may clear that list. That limit is acceptable for this assignment.

4. Wireframes — catalog and form

Musanze Safe Markets
MOB-G03-7014

Search stalls

Kinigi Produce Row

Produce · High

Open

Muhoza Grain Shed

Grains · Medium

Pending

Cyuve Meat Corner

Meat · High

Flagged

Home

New Inspection

Records

Home. Header holds the group code. Search filters the six cards. Cards are one reusable component. Status is a word inside a chip, not colour alone. Priority is written out.

Musanze Safe Markets
MOB-G03-7014

Search: zzzzz

No stalls match

Nothing in today's pilot uses that name.

Clear search

Home

New Inspection

Records

Empty state. Reachable by typing a search with no match. The message names the cause. Clear search restores the six cards. This is how test 2 shows the empty state.

Musanze Safe Markets
MOB-G03-7014

Vendor alias

Enter at least 2 characters.

Stall code · MZ-A-014

Use MZ-A-014 style: MZ, A-F, three digits.

Category

+250 7__ __ __

Use +250 7XX XXX XXX.

Risk · Low / Medium / High

☐ Fictional data consent

Continue to review

Home

New Inspection

Records

Form. Every error sits under its own field. The button stays, but it does not navigate while anything is invalid. Helper text shows one correct example.

5. Wireframes — image, review, records

Evidence photo

Preview of the selected stall photo.

Replace photo

Remove photo

If denied: Camera access is off. Try again, or choose a gallery photo.

If cancelled: No image selected. You can try again.

Image states. Empty, preview, replace, remove, denied, cancelled. Denied and cancelled never wipe the other form fields.

Review inspection
MOB-G03-7014

Mama Keza
MZ-A-014 · Produce
+250 788 123 456
Risk: High
Consent: Yes

Saved at 30 Sep 2026, 14:05

Save inspection

Back to edit

Review. Shows every validated field, the image, the timestamp, and the group code again. Save is the only way into the Records list.

Records
MOB-G03-7014

Mama Keza
MZ-A-014 · High
30 Sep 2026, 14:05

Detail
Same fields, photo, time, group code.
Back returns here.

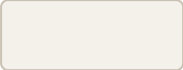
Home New Inspection Records

Records and detail. Records is a tab. Tapping a row pushes InspectionDetail. The list is still there when the officer comes back. An empty Records list says “No inspections saved in this session.”

6. Visual system

The palette is a deep market green on warm paper, chosen so text stays readable outdoors. Colour supports the label. It never replaces it.


Primary
#1B4D3E


Background
#F4F1EA


Surface
#FFFFFF


Text
#1A1A1A


Muted
#5C574E


Accent
#C45C26


Open
#1B6B3A


Pending
#8A5A00


Flagged / error
#9B2335

Type

Use the phone’s system font so the Android font-scale test can enlarge it. Do not lock text in fixed-height boxes.

Role	Size at default scale	Weight	Use
Screen title	22	700	Home, form, review titles
Card title	17	700	Stall name, vendor alias
Body	16	400	Fields, helper text, review rows
Meta	14	400	Category, time, group code line

Spacing and touch

- Screen padding 16. Gap between cards 12. Card padding 12.
- Buttons and list rows at least 48 high. Gap between two buttons 8.
- Corner radius 8 on cards and buttons. Chips are fully rounded.
- One filled primary button per screen. Secondary actions are outlined.

7. Components

Component	Looks like	Rules
App header	Green bar, white title, group code under the title	Present on every tab and on Review
Stall card	White card, grey placeholder, name, category, priority word, status chip	Same component for all six zones
Status chip	Open green, Pending amber, Flagged red, Closed grey	Text is always visible. Border plus fill.
Text field	White box, 1 px border, label above, error below in Flagged red	Controlled input. Error clears when that field becomes valid.
Choice row	Category list and Low / Medium / High	Selected row uses primary border and a written "Selected"
Primary button	Filled #1B4D3E, white label	Disabled look only after a failed attempt is not enough: explain the errors
Photo block	Full-width preview, Replace, Remove	Empty state offers Capture and Choose from gallery

Final mockups are these wireframes with the colours, type, and spacing on page 6 applied. Do not invent a second visual style in code. If a screen in the app disagrees with this document, change the app or update this PDF in the same commit so they still match.

Home at a glance. Warm background, green header with MOB-G03-7014, search field, six white cards. High-priority stalls are not a different layout. Priority is the word High, Medium, or Low so a quick scan still works when colours wash out in sun.

Review at a glance. Same header. A definition list of the six answers, the photo, the local timestamp, then Save inspection. The group code is in the header and repeated beside the timestamp.

8. Accessibility and three decisions

- **Contrast.** Body text #1A1A1A on #FFFFFF and on #F4F1EA. Primary button is white on #1B4D3E. Error text #9B2335 is also paired with the words of the message.
- **Touch targets.** Buttons, tabs, cards, and the consent row are at least 48 by 48.
- **Font scale.** Layout is Flexbox with wrapping text. No clipped single-line chips. The assessor will enlarge the device font.
- **Errors.** Each message names the field and the expected pattern. Example: "Stall code must look like MZ-A-014."
- **Not colour alone.** Status chips include the word Open, Pending, Flagged, or Closed.

Decision 1 — Search is how the empty state appears

The pilot always has six assigned stalls, so a permanently empty home would be false. A search box lets the officer narrow the list and lets the assessor reach a real empty state. Clear search puts the six stalls back. Justified because test 2 asks for six records and an empty state in the same catalog.

Decision 2 — The photo is required before Review

The scenario says the officer documents the stall with one evidence image, and the review screen must show that image. The form therefore blocks Review until a preview exists. Permission denial and cancellation stay on the form with a recovery message, so a refused camera does not trap the officer: they can try again or use the gallery.

Decision 3 — One exact stall-code pattern, shown on the field

The brief asks for an exact stall code pattern but does not print one. The product uses MZ-[A-F]-NNN so every zone letter in the catalog (A through F) can match a stall, and the officer can see a correct example under the field. The same pattern is what Member 3 validates. Unseen wrong codes such as A14, mz-a-14, or MZ-G-001 fail.

Member 1 evidence: this document is the owned design. Implementation must follow design/SCREEN_SPEC.md. Export this file as MOB_A1_G03_UIUX.pdf.